

- 11 (a) sharpness: ease with which edges of structures can be seen B1
 contrast: difference in degree of blackening between structures B1 [2]
- (b) (i) $I = I_0 e^{-\mu x}$ C1
 $I/I_0 = \exp(-0.20 \times 8)$
 $= 0.20$ A1 [2]
- (ii) $I/I_0 = \exp(-\mu_1 \times x_1) \times \exp(-\mu_2 \times x_2)$ (could be three terms) C1
 $I/I_0 = \exp(-0.20 \times 4) \times \exp(-12 \times 4)$ C1
 $I/I_0 = 6.4 \times 10^{-22}$ or $I/I_0 \approx 0$ A1 [3]
- (c) (i) sharpness unknown/no B1 [1]
- (ii) contrast good/yes (ecf from (b)) B1 [1]

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12 (a) ω carrier frequencies can be re-used (without interference)

(M1)